PLANT INSPECTION REPORT

Inspection & Integrity Department — Field Record

Report No.:	INS-2026-0417	**Equipment Tag:**	V-2104
Description:	Crude Overhead Knock-Out Drum	**Service:**	Sour hydrocarbon vapour / condens
Design Pressure:	10.5 bar(g)	**Design Temp.:**	145 deg C
Material:	SA-516 Gr.70 Carbon Steel	**Nominal Thickness:**	12.0 mm
Date of Inspection:	18 February 2026	**Previous Inspection:**	20 February 2022
Inspector:	R. Menon (Cert. API-510 #38812)	**Method:**	UT thickness survey + visual exte

ULTRASONIC THICKNESS READINGS (mm)

Location	2022	2026	Min. Recorded
Shell course 1 (top)	11.8	10.9	10.9
Shell course 2 (mid)	11.6	9.4	9.4
Shell course 3 (bot)	11.9	11.1	11.1
Bottom head	12.0	11.4	11.4
Inlet nozzle N1	11.5	10.2	10.2
Manway M1 flange	12.0	11.8	11.8

VISUAL OBSERVATIONS

1. External cladding damaged over approx. 35% of shell course 2; insulation found waterlogged on removal. Rust bleeding at the cladding joints on the north face.
2. Localised external metal loss beneath damaged insulation on shell course 2. Pitting observed, max pit depth 1.6 mm, density approx. 12 pits per 100 sq.cm.
3. Inlet nozzle N1 shows general wall loss consistent with erosion-corrosion at the inlet impingement zone.
4. Relief valve PSV-2104A tested satisfactory on 12 Jan 2026.
5. Support saddles, earthing and nameplate in good condition.
6. No through-wall defects or active leaks identified.

INSPECTOR REMARKS

Corrosion under insulation confirmed on shell course 2. Wall loss at course 2 is the governing location. Recommend engineering assessment of remaining life before the next operating campaign, and repair of cladding and insulation during the current shutdown window. Vessel considered fit for continued service in the interim, subject to review.

Signed: R. Menon Date: 18/02/2026